PVsyst V8.0.20

PVsyst - Simulation report

Grid-Connected System

Project: 22 kWp Grid-Connected Rooftop PV System — Kömürcüler, Antalya

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 22.00 kWp

Kömürcüler - Turkey

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Author

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Project summary

Geographical Site
Kömürcüler
Turkey

Situation

Latitude	37.07 °(N)
Longitude	30.62 °(E)
Altitude	297 m
Time zone	UTC+3

Project settings

Albedo	0.20

Weather data
Kömürcüler
Meteonorm 8.2 (2003-2013), Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Fixed plane
Tilt/Azimuth 7 / 0 °

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	40 units
Pnom total	22.00 kWp

Inverters

Nb. of units	1 unit
Total power	20 kWac
Pnom ratio	1.10

Results summary

Produced Energy	31613 kWh/year	Specific production	1437 kWh/kWp/year	Perf. Ratio PR	84.29 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Main results 4
Loss diagram 5
Predef. graphs 6
Single-line diagram 7

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Orientation #1
Fixed plane
Tilt/Azimuth 7 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	TSM-DE19-550
(Original PVsyst database)	
Unit Nom. Power	550 Wp
Number of PV modules	40 units
Nominal (STC)	22.00 kWp
Modules	4 string x 10 In series
At operating cond. (50°C)	
Pmpp	20.13 kWp
U mpp	286 V
I mpp	70 A

Inverter

Manufacturer	Generic
Model	SUN2000-20KTL-M2
(Original PVsyst database)	
Unit Nom. Power	20.0 kWac
Number of inverters	1 unit
Total power	20.0 kWac
Operating voltage	160-950 V
Max. power (=>45°C)	22.0 kWac
Pnom ratio (DC:AC)	1.10
Power sharing within this inverter	

Total PV power

Nominal (STC)	22 kWp
Total	40 modules
Module area	105 m²

Total inverter power

Total power	20 kWac
Max. power	22 kWac
Number of inverters	1 unit
Pnom ratio	1.10

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 67 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.40 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.05 %

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000

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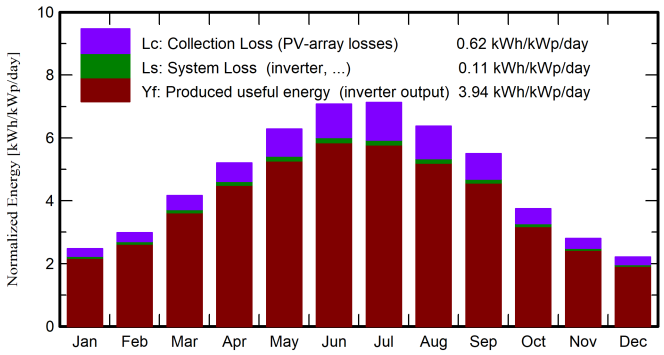
Main results

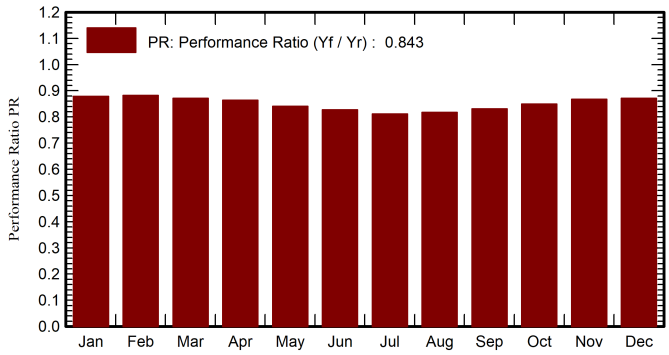
System Production

Produced Energy	31613 kWh/year	Specific production	1437 kWh/kWp/year
		Perf. Ratio PR	84.29 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	67.0	28.71	8.84	76.8	72.5	1529	1485	0.879
February	76.5	40.47	9.98	83.6	79.7	1667	1621	0.882
March	121.2	55.42	12.67	129.0	124.2	2542	2472	0.871
April	151.2	75.22	15.59	156.2	150.8	3050	2969	0.864
May	192.3	74.95	20.28	194.7	188.7	3702	3601	0.841
June	211.7	83.41	24.80	212.3	205.9	3971	3866	0.828
July	219.3	73.73	28.49	220.9	214.1	4051	3942	0.811
August	191.9	76.08	28.65	197.5	191.7	3648	3550	0.817
September	155.4	57.06	24.44	165.0	159.1	3096	3015	0.831
October	107.1	50.30	19.83	116.3	111.6	2234	2174	0.850
November	74.9	37.63	14.20	84.2	79.8	1650	1605	0.867
December	59.5	29.17	10.24	68.5	64.3	1352	1312	0.871
Year	1627.9	682.14	18.22	1704.7	1642.6	32492	31613	0.843

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

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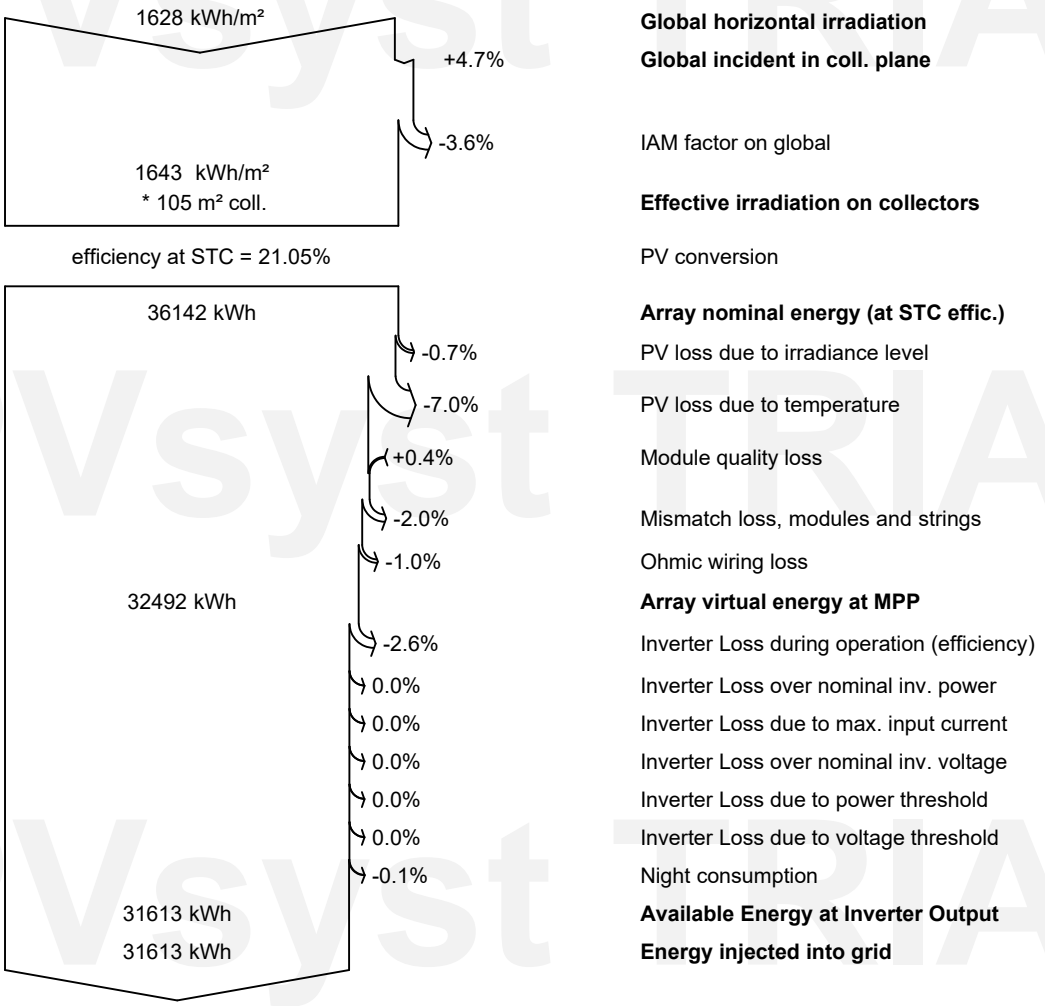
Loss diagram



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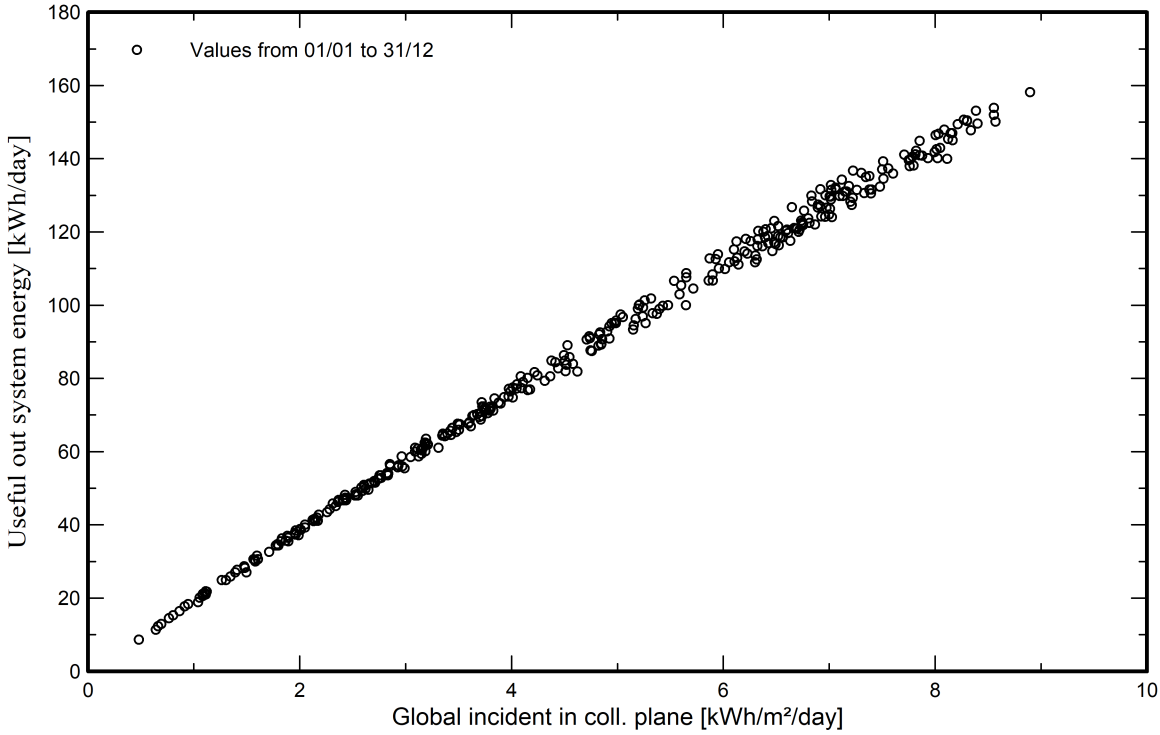
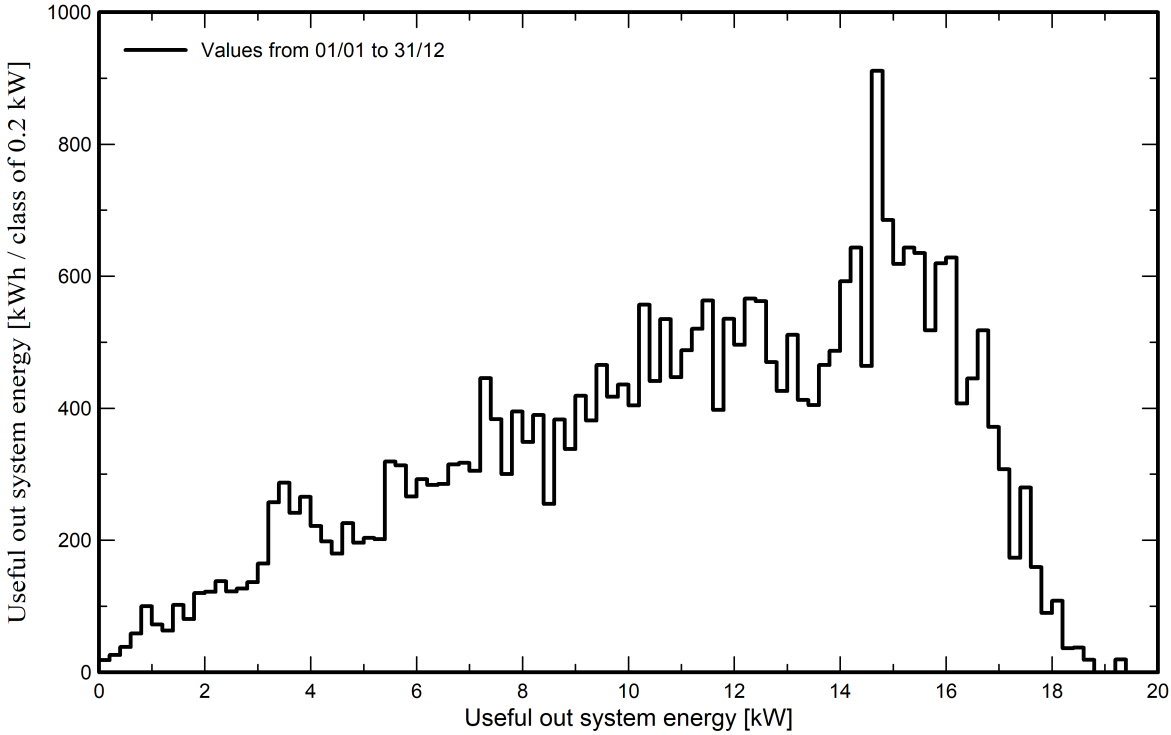
Predef. graphs

Daily Input/Output diagram

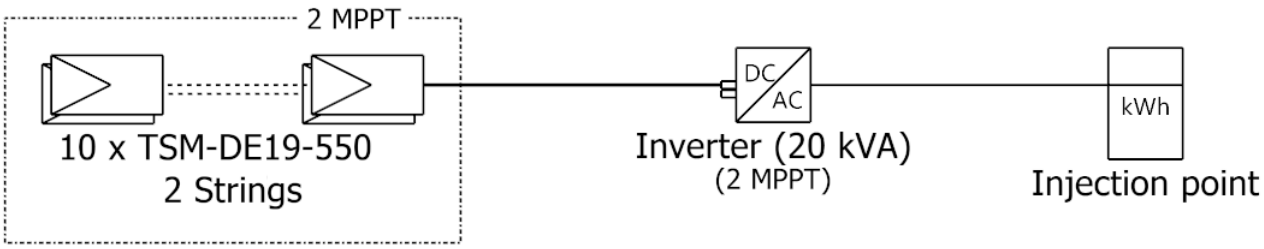


System Output Power Distribution



Single-line diagram

PVsyst V8.0.20
VC0, Simulation date:
08/03/26 15:08
with V8.0.20



PV module	TSM-DE19-550
Inverter	SUN2000-20KTL-M2
String	10 x TSM-DE19-550

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VC0 : New simulation variant

08/03/26